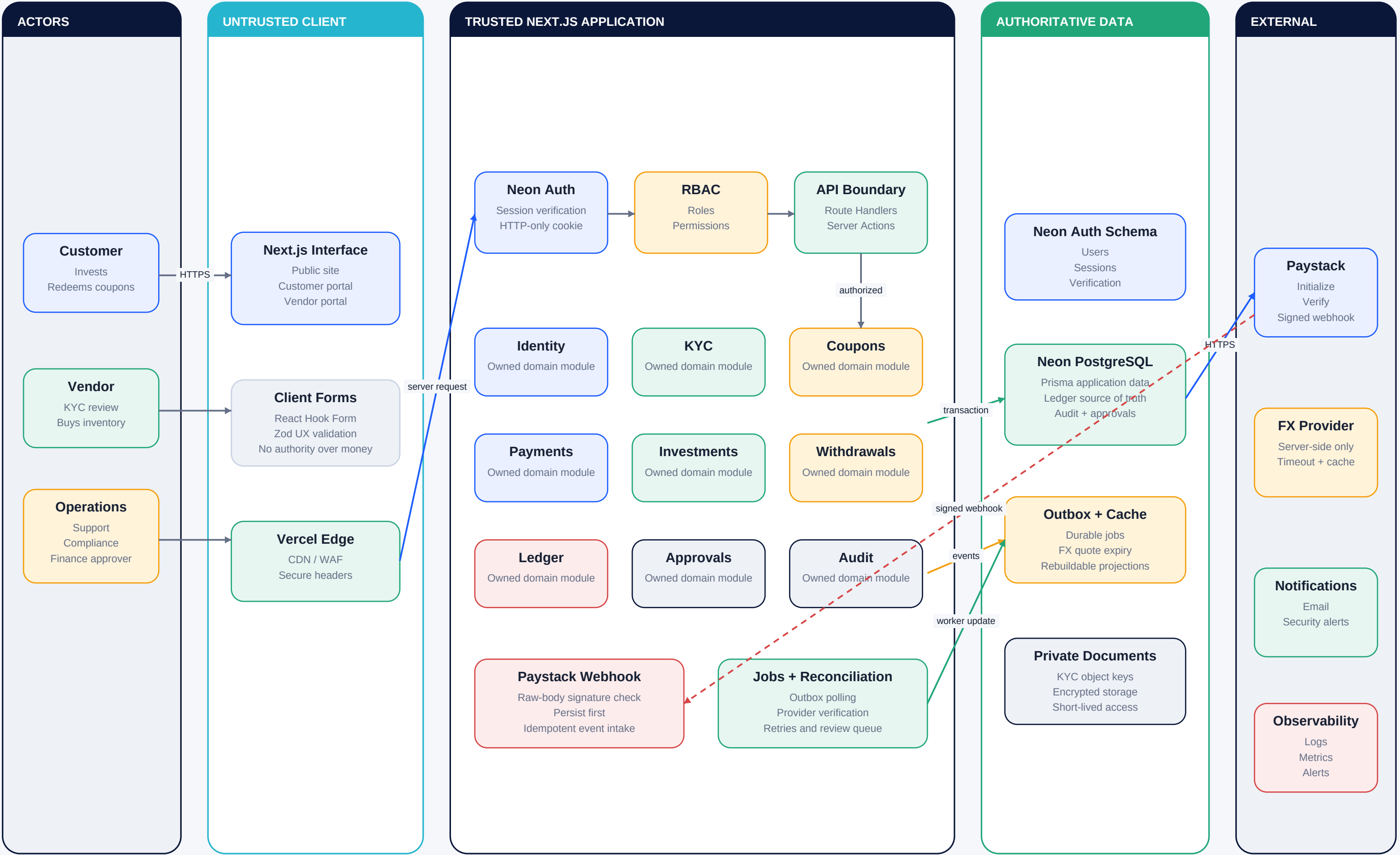
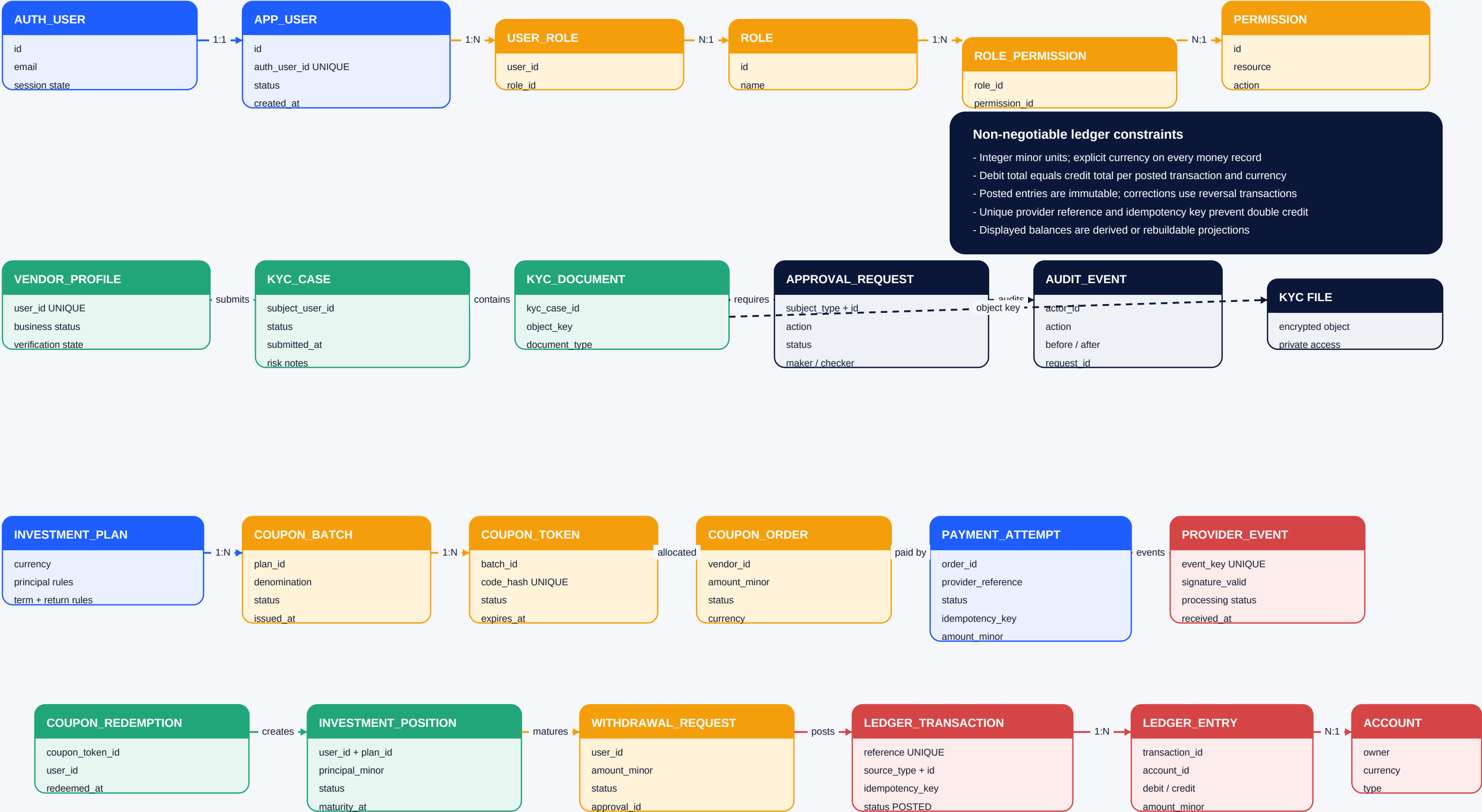


Great Finance - Unified Production Architecture

End-to-end trust boundaries, runtime components and authoritative data flow





Paystack Payment and Ledger Posting

Signed webhook intake, durable processing, provider verification and exactly-once economic effect



Never credit from the browser callback, URL amount, screenshot or webhook payload alone.

Require valid HMAC signature, unique event intake, Paystack verification, exact amount/currency/reference match and one balanced ledger transaction.



RBAC + MAKER-CHECKER

Role examples

Customer / Vendor
Support / Compliance
Finance / Admin

Enforcement

Session + permission
Resource ownership
Server boundary only

Approval rules

Requester cannot approve own operation
Large payouts may require two approvers
Reasons and evidence are mandatory

SECURITY + OPERATIONS

- No passwords or provider secrets in application tables

- Zod validation at every command boundary

- Rate limits, secure headers and CSRF-aware mutations

- Structured logs without sensitive values

- Alerts for webhook failures and ledger mismatches

- Neon branch migration tests and restore drills

- Runtime pooled URL; controlled migration connection

- Private KYC files with short-lived access

DELIVERY ORDER

1

Security foundation
Remove Firebase secrets/password storage; establish Auth + RBAC

2

Database foundation
Prisma schema, migrations, audit, approvals and ledger

3

Payments
Paystack verification, webhook inbox, outbox and reconciliation

4

Product workflows
Coupon inventory, redemption, investment and withdrawal

5

Production hardening
Tests, monitoring, accessibility, SEO and restore drills

Production gate

Release only after authorization tests, concurrent ledger tests, webhook replay tests, Paystack test-mode reconciliation, migration rollback, Neon restore drill, audit-log review, responsive browser verification and measured load tests all pass.

The diagram defines the target architecture; scalability is confirmed by production-like evidence, not by the diagram alone.